Capacitor Selection and Reliability Optimization System for a DC–DC Converter

MATLAB-Based Research Framework with Commercial Component Screening and Simulink Validation Plan

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Abstract— This report presents a staged framework for selecting commercially available output capacitors for a closed-loop buck converter. The prototype combines analytical ripple-current calculations, tolerance-aware capacitance screening, ESR-related ripple and loss estimates, a ten-part commercial shortlist, strict qualification flags, and a weighted scoring architecture. The baseline is a 24 V input, 12 V output reference, 1 mH inductor, 50 kHz switching frequency, and 3.75 Ω load. Calculations yield an ideal duty ratio of 0.5, 0.12 A peak-to-peak inductor ripple, approximately 34.6 mA capacitor RMS ripple current, and 3 µF ideal minimum capacitance for a 100 mV capacitive-ripple target. A nominal 22 µF candidate has 17.6 µF minimum capacitance under a ±20% tolerance. The report separates provisional estimates from verified manufacturer ratings and does not declare a final winning part because several ratings, life conditions, costs, and thermal data remain incomplete.

Index Terms— capacitor selection, buck converter, ESR, ripple current, reliability, lifetime estimation, multi-criteria optimization, MATLAB, Simulink.

I. INTRODUCTION

Output capacitors influence ripple, load-step response, thermal stress, control-loop behavior, footprint, cost, and service life. Selecting by nominal capacitance and voltage alone is insufficient. This project frames capacitor selection as constrained optimization with data provenance and validation. The initial application is the supplied buck converter; future profiles may cover buck–boost converters, DC links, filters, motor drives, and IC decoupling.

II. OBJECTIVES AND SCOPE

The goals are to derive application-specific electrical requirements; maintain a traceable commercial database; reject hard-constraint failures; estimate ripple and ESR loss when data exist; distinguish provisional scores from strict qualification; and define repeatable Simulink/hardware validation. First-pass equations assume ideal continuous-conduction operation and omit switch drops, ESL, PCB parasitics, control-loop dynamics, thermal coupling, and DC-bias effects.

III. SYSTEM MODEL AND BLOCK DIAGRAM

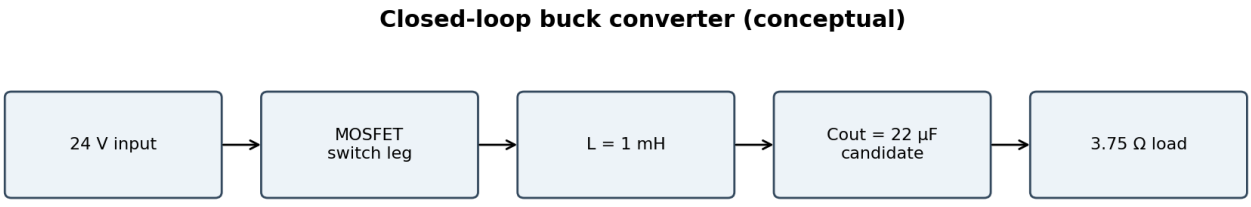


Fig. 1. Conceptual closed-loop buck converter and output capacitor under test.

The baseline model uses a 24 V source, MOSFET switching stage, 1 mH inductor, output capacitor, 3.75 Ω load, 12 V reference, PI controller, and 50 kHz PWM (20 µs period). The cyclic-load profile must be identical for every candidate. A buck–boost extension reuses the database architecture but requires topology-specific equations and stress constraints.

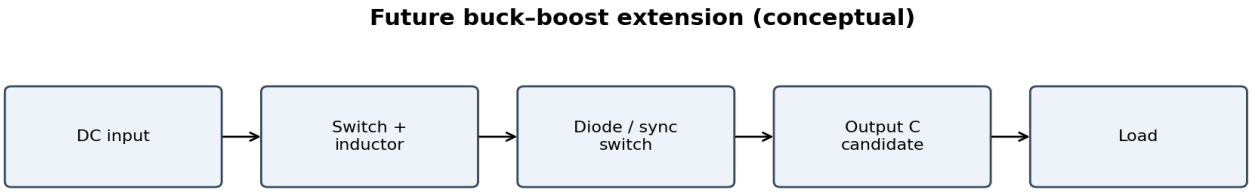


Fig. 2. Proposed buck–boost extension; not the presently evaluated topology.

IV. PARAMETER CALCULATIONS

The ideal duty ratio is D = Vout/Vin = 12/24 = 0.500. Inductor ripple is ΔIL = (Vin − Vout)D/(Lfs) = 0.120 A peak-to-peak. Approximate capacitor RMS ripple current is IC,rms ≈ ΔIL/(2√3) = 0.0346 A. The fixed-load current is Vout/Rload = 3.20 A, and output power is 38.4 W.

The ideal capacitive ripple relation is ΔVC ≈ ΔIL/(8fsC). For a 100 mV target, the ideal lower bound is C = 3.00 µF. A nominal 22 µF capacitor with ±20% tolerance has Cmin = 17.6 µF, giving ideal capacitive ripple of 17.05 mV p-p. This lower bound does not include transient, stability, thermal, DC-bias or practical margin requirements.

Parameter	Equation / basis	Result
Duty ratio	Vout/Vin	0.500
Inductor ripple	(Vin–Vout)D/(Lfs	0.120 A p-p
Capacitor RMS ripple	ΔIL/(2√3)	34.6 mA
Output current	Vout/Rload	3.20 A
Output power	Vout × Iout	38.4 W
Ideal capacitance lower bound	ΔIL/(8fsΔVC), ΔVC=0.1 V	3.00 µF
22 µF tolerance minimum	22 × 0.8	17.6 µF
Capacitive ripple at Cmin	ΔIL/(8fsCmin)	17.05 mV p-p

V. COMMERCIAL CAPACITOR SHORTLIST

The following ten part numbers are a research shortlist, not ten fully verified interchangeable parts. Panasonic 35SVPF22M is manufacturer-documented at 22 µF, 35 V, 35 mΩ ESR at 100 kHz, 2.6 A RMS ripple at 100 kHz and 105°C, and 5000 h endurance at 105°C [1]. Panasonic EEE-HB1V220AP is 22 µF, 35 V, ±20%, and 55 mA at 120 Hz; its ESR is not listed in the product table [2]. KEMET T521V226M025ATE045 is listed at 45 mΩ at 100 kHz and 2 A ripple current [3]. Other entered values remain provisional until exact datasheets and test conditions are confirmed.

TABLE I. TEN COMMERCIAL CANDIDATES (PROVISIONAL)

Manufacturer	Part number	V	ESR mΩ	Ripple A	Data-quality note
Panasonic	EEE-HB1V220AP	35	Unknown	0.055	55 mA at 120 Hz; ESR not published in product table
Panasonic	25TQC22MD2	25	60	1.4	Prior listing; verify exact datasheet conditions
Panasonic	35SVPD22M	35	50	2.3	Prior listing; verify exact datasheet conditions
KEMET	T521V226M025ATE045	25	45	2.0	45 mΩ at 100 kHz; 2 A distributor listing
KEMET	T521B226M025ATE100	25	100	1.1	Prior listing; verify exact datasheet conditions
KEMET	A758EK226M1EAAE050	25	50	2.0	Prior listing; verify exact datasheet conditions
Nichicon	FP-025ME220M-SSR	25	40	2.1	Prior listing; verify exact datasheet conditions
Panasonic	35SVF22M	35	35	0.82	Verify exact ripple-current conditions
Panasonic	35SVPF22M	35	35	2.6	Official data: 2.6 A at 100 kHz, 105°C; 5000 h at 105°C
Panasonic	EEE-FN1V220R	35	700	0.16	Prior listing; verify exact datasheet conditions

VI. ALGORITHM EVOLUTION: V1–V4

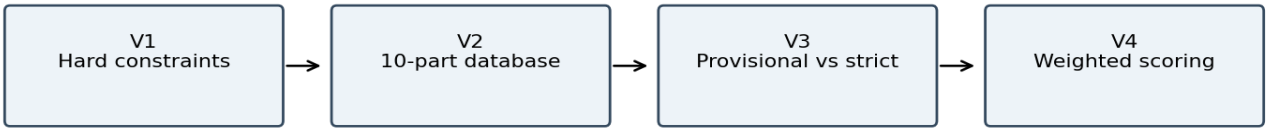


Fig. 3. Algorithm development from hard screening to weighted scoring.

V1 — Constraint screening: calculate duty ratio, inductor ripple, capacitor RMS current and ideal capacitance; check voltage/capacitance constraints. It is transparent but cannot rank parts with different ESR, life, price or data quality.

V2 — Commercial database and ESR estimates: add ten exact part numbers and estimate ESR ripple and dissipation where ESR is known. Unknown values remain NaN. A strict qualification flag is introduced.

V3 — Provisional vs. strict qualification: provisional screening identifies candidates for further investigation; strict qualification additionally requires verified datasheet conditions. All verification flags currently remain false.

V4 — Weighted scoring: add a 0–100 electrical score and placeholders for reliability, lifetime and cost. The final score remains NaN until all criteria are known and strict constraints pass.

VII. SCORING METHODOLOGY

The proposed weights are electrical performance 30%, reliability 30%, lifetime 25%, and cost 15%. The electrical subscore uses 70% ripple and 30% ESR loss. For a lower-is-better metric x, relative normalization is $S_x = 100(x_{max}-x)/(x_{max}-x_{min})$. This is relative to the current candidate set, not an absolute quality or failure-probability metric. Reliability requires verified current ratings; lifetime requires a manufacturer-specific life model and operating temperature; cost requires comparable price, quantity, currency and date.

Criterion	Weight	Evidence needed
Electrical	30%	Ripple and ESR-loss estimates
Reliability	30%	Verified ripple-current margin
Lifetime	25%	Manufacturer life model and operating temperature
Cost	15%	Comparable dated price and quantity

VIII. PRELIMINARY RESULTS AND RANKING

Using the current entered ESR values, $\Delta V_{ESR} \approx \Delta I_L \cdot ESR$, $P_{ESR} \approx I_{C,rms}^2 \cdot ESR$, and total ripple estimate is $\Delta V_C + \Delta V_{ESR}$. The lowest estimate is about 21.245 mV p-p for Panasonic 35SVF22M and 35SVPF22M, followed by 21.845 mV for Nichicon FP-025ME220M-SSR. These values are preliminary estimates, not measured Simulink results.

Part number	ESR mΩ	Ripple mV p-p	Loss µW	Electrical score
35SVF22M	35.0	21.25	42.00	100.0
35SVPF22M	35.0	21.25	42.00	100.0
FP-025ME220M-SSR	40.0	21.85	48.00	99.2
T521V226M025ATE045	45.0	22.45	54.00	98.5
35SVPD22M	50.0	23.05	60.00	97.7
A758EK226M1EAAE050	50.0	23.05	60.00	97.7
25TQC22MD2	60.0	24.25	72.00	96.2
T521B226M025ATE100	100.0	29.05	120.00	90.2
EEE-FN1V220R	700.0	101.05	840.00	0.0

The current MATLAB output reports nine provisional passes and zero fully qualified parts. This is expected because all RatingConditionsVerified flags are false. EEE-HB1V220AP cannot receive an ESR-based ripple estimate because ESR is unknown. EEE-FN1V220R exceeds the 100 mV estimated ripple threshold using the entered 0.7 Ω ESR.

IX. OUTPUT RESPONSE

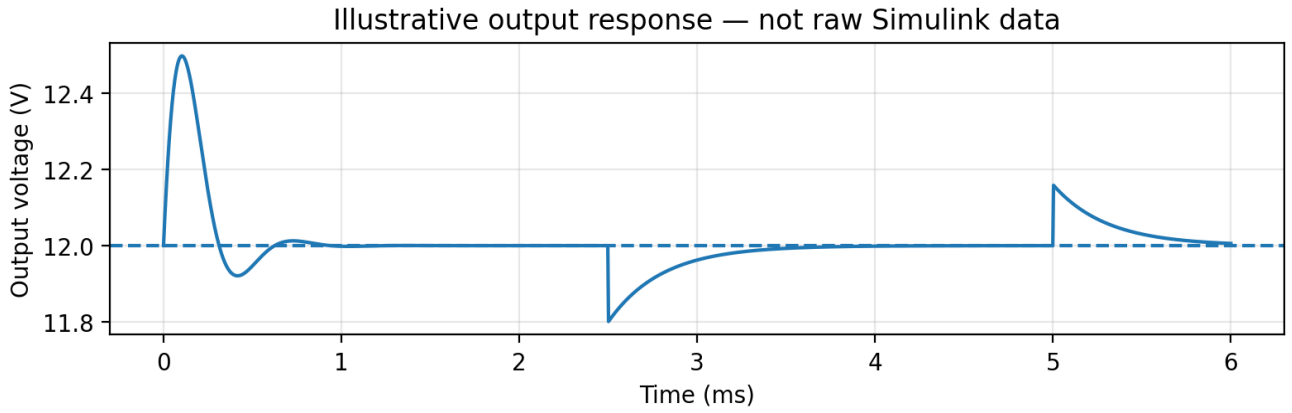


Fig. 4. Constructed illustrative response, not raw Simulink data.

The user-provided screenshots qualitatively showed startup overshoot near 13.06 V, settling near 12 V, and cyclic-load disturbances around 2.5 ms and 5 ms. These are cursor readings from screenshots rather than exported simulation data. The figure is deliberately labelled illustrative and must not be cited as a measured result.

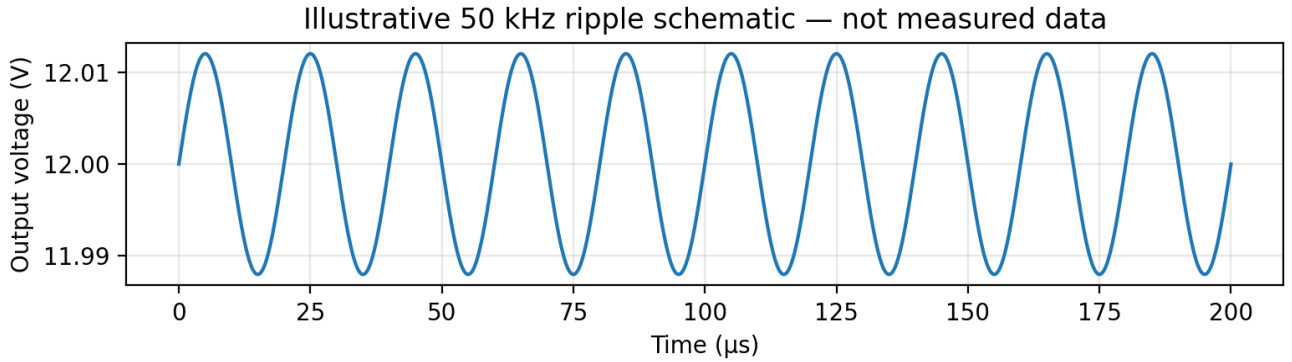


Fig. 5. Illustrative 50 kHz ripple schematic; not measured output ripple.

To measure actual ripple, export Scope data or enable signal logging and calculate max(Vout)−min(Vout) over several settled switching periods. At 50 kHz each period is 20 µs. Do not confuse startup transient values with switching ripple.

X. SIMULINK VALIDATION PLAN

Test	Procedure	Metrics
Steady state	Capture several settled PWM cycles	Vout p-p ripple; mean
Startup	Same initial conditions	Overshoot; settling time
Load increase	Same load step for all parts	Undershoot; recovery time
Load decrease	Same load removal	Overshoot; recovery time
Thermal estimate	Use verified ESR and RMS current	PESR; temperature rise
Sensitivity	Sweep C tolerance, ESR, load, temperature	Worst-case margin

Keep the converter, controller, solver, initial conditions, load profile and measurement window identical. Change only capacitor parameters. If ESR is unknown, do not substitute zero; obtain data, run a documented sensitivity range, or exclude that part from ESR-based ranking.

XI. LIMITATIONS AND ENGINEERING RISKS

ESR varies with frequency and temperature. Ripple ratings may use different test conditions. The additive ripple expression is a first-order estimate, not a full impedance-domain simulation. Lifetime models vary by technology. Price and life inputs are missing for most parts. Confirm package, polarity, derating, surge/inrush requirements and current manufacturer datasheets before procurement. The 100 mV target is a screening threshold selected for this prototype, not a universally valid design requirement.

XII. RESEARCH ROADMAP

1) Verify every part-number record and retain source URL, datasheet revision, retrieval date, test frequency, temperature, package, tolerance, life and price. 2) Add frequency-dependent impedance/ESR curves and thermal models. 3) Add tolerance/temperature sensitivity or Monte Carlo analysis. 4) Automate Simulink sweeps and metric extraction. 5) Validate selected candidates on hardware using an oscilloscope and temperature measurement. 6) Extend to buck–boost, DC-link, filters, motor drives and IC decoupling.

XIII. CONCLUSION

This prototype combines transparent converter calculations, a ten-part commercial shortlist, provisional electrical ranking and strict qualification gates. It does not yet establish a final winner: nine candidates pass provisional screening, but zero are fully qualified because rating conditions are unverified. Datasheet provenance and repeatable simulation measurements are the next essential steps.

XIV. REPOSITORY AND REPRODUCIBILITY

A GitHub repository has not been created from this chat. The source package includes MATLAB code, report, diagrams and README for upload. Suggested repository name: capacitor-selection-optimization-system. After creating a repository under your account, replace **https://github.com/<your-username>/capacitor-selection-optimization-system** with its actual URL. This is a placeholder, not an active repository link.

REFERENCES

[1] Panasonic Industry, “35SVPF22M: Conductive Polymer Aluminum Solid Capacitors (OS-CON),” https://industrial.panasonic.com/ww/products/pt/os-con/models/35SVPF22M
[2] Panasonic Industry, “EEEHB1V220AP: Aluminum Electrolytic Capacitors (Surface Mount Type),” https://industrial.panasonic.com/ww/products/pt/aluminum-cap-smd/models/EEEHB1V220AP
[3] KEMET T521V226M025ATE045 product/datasheet information, DigiKey, https://www.digikey.in/en/products/detail/kemet/T521V226M025ATE045/5131700
[4] IEEE Author Center, conference template resources, https://template-selector.ieee.org/
[5] MathWorks, MATLAB table documentation, https://www.mathworks.com/help/matlab/tables.html
[6] MathWorks, Simulink documentation, https://www.mathworks.com/help/simulink/